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2nd International Workshop on Hybrid Collaboration–Analyzing Collaborative Interaction

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Abstract. Hybrid collaboration, where co-located participants are working together with remote participants, is increasingly established as a de-facto standard practice in our everyday professional lives. Since the first edition of this workshop in 2019, the pool of research has considerably increased. However, studying such collaborative practices is still met with skepticism by researchers due to the considerable effort that is connected to the according empirical endeavors. In this workshop, we aim at bringing together researchers and practitioners who are interested in the analysis of collaborative practices in general (i.e., co-located or remote), and/or a domain transfer to hybrid collaboration in particular.

Workshop Themes

In recent years, hybrid collaboration has climbed the rungs of our meeting culture to be one of the major settings in which we work together, next to traditional face-to-face meetings and purely remote collaboration. On Johansen (1988)'s traditional time-space matrix, all four quadrants can be dynamically traversed in

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such hybrid situations: Both **co-located** and **remote** participants might engage in both **synchronous** and **asynchronous** work. Therefore, as hybrid collaboration encompasses more complex and ubiquitous interaction among team members as well as their personal and shared devices regardless of the differences in time and space, new opportunities and challenges emerge for CSCW researchers in how to observe, analyze, evaluate, and prototype hybrid collaboration. Jordan (2009) mentioned the obscurity of borders between the real and virtual in hybrid spaces and underlined the need for conducting hybrid ethnography, which incorporates research in both virtual and physical spaces. In other words, as Prior and Miller (2012) also pointed out, while the rise of online and virtual platforms and tools paved the path for the development of new methods as well as discussions of virtual/online ethnography or webethnography, these approaches are valid only in observing the online-only communities and are not always applicable in analyzing both online and offline interactions around those platforms and tools within a group of people. Understanding and evaluating hybrid collaboration at a business setting today requires the triangulation of various qualitative and/or quantitative research methods, which could allow the researchers to make sense of the overall hybrid experiences of collaboration rather than reproducing the **virtual** *or* **real** dichotomy. Another dichotomy was a major focal point in the first workshop in 2019, where Neumayr et al. (2019) made the case that "CSCW research has traditionally focused on different styles of group work and has emphasized the distinction of either" studying **co-located** *or* **remote** collaboration in isolation and rarely hybrid practices were explicitly considered (also see Neumayr et al. (2022c)). While this has arguably changed for the better in recent years, the wealth of traditional models and frameworks for the description, analysis and according support of hybrid teams, could barely keep up with practical applications. For instance, as outlined in Neumayr et al. (2019) for the remote tradition "Gutwin and Greenberg (1998) developed the notion of *workspace awareness*" and "coined the term *mixed-focus collaboration*" (Gutwin and Greenberg, 2002), later O'Hara et al. (2011) investigated *blended interaction spaces*. In the co-located tradition, Tang et al. (2006) and Isenberg et al. (2012) developed their frameworks of *coupling styles*, and Scott et al. (2004) discovered the usefulness of the phenomenon *territoriality* for the analysis of co-located collaboration.

While there have been efforts to extend these analytical lenses to be able to capture hybrid collaborative practices, such as Neumayr et al. (2018)'s hybrid coupling styles and Neumayr et al. (2022a)'s approach to make territoriality hybrid-fit, more work has to be done. In CSCW research as a socio-technical field, the core items of research interest are people interacting with one another as well as with technology. Numerous idiosyncratic coding schemes have been developed over the years, but their lowest common denominator is that when analyzing observational data (as a great supplement to self-reported data, e.g., through interviews or questionnaires) after a study was conducted and has to be analyzed, the relations between people, objects, and technology have to be considered. On the other hand, collaborative practices of participants in hybrid configurations have

also drawn an increasing amount of attention in other research fields, of which one is social interaction and conversation analysis (CA). CA is a bottom-up empirical approach to the study of human social conduct that has its roots in the 1960's sociology. CA scholars use audio or video recordings of authentic situations as the main research data with the aim to unravel how participants in interaction use talk and multimodal resources (e.g., gaze, gestures, and material and digital objects) to advance interaction and establish intersubjective understanding, then and there (e.g., (Mondada, 2014); see also (Sidnell and Stivers, 2012)). With its focus on *situated social actions*, CA can provide valuable insights into the ways in which participants in hybrid configurations establish and sustain the (pre)conditions for interaction, or their shared *interactional space* (see (Oittinen, 2020b)), develop practices for collaboration, and solve troubles related to it (Oittinen, 2020a). A notable example of CA methodology being used to make sense of hybrid collaboration is described in (Saatçi et al., 2020).

The issues raised until here show that there are still many open questions resulting in a knowledge gap that entails a broad spectrum of hybrid collaboration topics. To address some of these aspects and besides the approach to make existing frameworks and models hybrid-fit described by Neumayr et al. (2018, 2022a) above, Saatçi et al. (2019) discussed the methodological challenges of observing and analyzing hybrid meetings in a business setting. The abundance of the participants and technical devices as well as the spatial complexity (i.e., the simultaneous co-construction of and orientation to "virtual" and "physical" spaces (Oittinen, 2020a)) makes it difficult for the researchers to make sense of hybrid collaboration and interaction in hybrid meetings and requires more in-depth contextual data collection. Yet, today, companies are workplaces with highly confidential data and restricted timeframes, which do not allow a far-reaching longitudinal study of the phenomenon.

Considerations like these explain the overall hesitance of researchers to conduct experiments of hybrid collaboration in general and even more so in real-world scenarios.

Now—as discussed in (Neumayr et al., 2022b)—all of the approaches mentioned above with either idiosyncratic coding schemes or detailed interaction analyses are more or less striving for objectivity and minimizing speculation while recording what can be seen and heard in audio-visual material. In all of these approaches, we would like to make sense of the interactions of individuals with others as well as with (technological) objects. In our opinion, such analysis endeavors bear great potential to be (at least partly) automated with the help of recent advances in artificial intelligence. Well-trained models could recognize and identify participants, locate them in a given spatial setting, make sense of their interactions, and ultimately could judge if their conversations are targeted at only a subgroup of a larger hybrid meeting or destined for the whole group. While we anticipate these imminent developments with optimism, numerous ethical questions remain: while it may be possible to do larger-scale and more naturalistic real-world studies of teams engaging in hybrid collaboration compared to what

human researchers and analyzers could achieve, privacy and confidentiality concerns remain when it comes to feeding recorded material to AI services. As discussed in Neumayr (2023) following Kranzberg (1986)'s first law, ultimately it is up to us as researchers to make responsible use of the capabilities and strive for real voluntarism of all involved stakeholders in our research endeavors.

In summary, we believe that there is a strong need for a methodological discussion among researchers studying hybrid collaboration regarding the problems and possible solutions, strategies or attempts of studying new collaborative practices and developing new multi-disciplinary mixed methods.

Thus, this workshop aims at bringing together researchers with expertise in CSCW scenarios including either remote collaboration or co-located settings to discuss possible common ground, how the findings of each field can benefit the other and find answers to the questions outlined above. Moreover, as previous years showed a surge in hybrid-focused research, scholars and practitioners directly engaged in the topic are especially invited to contribute their experiences in our workshop.

The topics dealt with in this workshop include but are not limited to:

- What are the opportunities and challenges of researching hybrid collaboration today?
- How can collaborative practices in hybrid meetings be studied in naturalistic environments, including methodology of different fields, such as CA (e.g., (Oittinen, 2018, 2020b, 2022; Saatçi et al., 2020; Büyükgüzel and Balaman, 2023))?
- What are the prospects of using automation in the analysis of recorded observational data of (hybrid) collaborative interaction?
- How can we prototype hybrid collaboration and conduct controlled experiments?
- Which established, alternative, or entirely novel research methods can help to tackle those challenges?
- For researchers who recently conducted inquiry of hybrid collaboration: What can we learn from prior (possibly recent) studies of hybrid collaboration; what are the major take-aways?

Workshop Activities and Goals

The workshop is planned not as a separate mini conference with conventional paper presentations, but as a mixture between short impulse presentations followed up by extensive discussion and dialog. For this purpose, an estimated number of 10 to 15 selected participants will team up to form small groups of three to four persons and work on current and future research agendas.

Goals

This full-day workshop aims at illuminating the intersections of co-located and remote collaboration research and mutually transferring results to the respective other field. There are notable examples of a successful transfer of findings from one area to the other, e.g., Tuddenham and Robinson (2009) who described phenomena for remote settings that were initially conceptualized for co-located collaboration, namely *territoriality* and *collaborative coupling styles*. Additionally, to further spur discussion, some aspects that are idiosyncratic to the aforementioned phenomena of hybrid collaboration meetings will be suggested by the organizers based on their prior research (e.g., the difficulty to conduct field studies of hybrid meetings in naturalistic environments or how to establish sufficient workspace awareness in a hybrid collaboration setting). A first step will be a discussion of the state of the art of both areas. Relevant topics which will be discussed can address i) an understanding of current collaborative practices, ii) the technological challenges involved in their execution, and iii) possible concepts that can be embedded into the architecture of future collaborative systems in order to ensure a high usability.

Before the workshop

The call for papers will be distributed over the involved institution's mailing lists and social networks. Additionally, relevant research groups (CSCW, HCI, interaction design, CA) will be addressed directly via personal invitations of acknowledged experts and their respective research groups.

Participants will be selected based on their position paper submissions (4 pages in length using the recent ACM SIGCHI Extended Abstracts format). The submissions will be reviewed by the workshop organizers and judged by their quality concerning relevance and potential to stimulate discussion at the workshop.

Accepted position papers will be distributed among the participants well before the workshop to allow a familiarization with each others' topics. In this process, participants are asked to reflect on the other submissions and bring questions with them.

At the workshop

A preliminary workshop schedule can be viewed in Table I. The workshop will span half a day and start with short impulse presentations of the participants' position papers. In the afternoon, there will be a discussion reflecting on the different presentations and possible connections between the topics. This is used as preparation to facilitate associative thinking which is helpful for the identification of current and future research agendas. Next, one or two research agendas are assigned to small groups of three to four participants who further reflect on them based on their own experiences and try to formulate a common research aim. During the concluding consolidation session, the different research

aims are discussed and rated based on their relevance. If possible, a common mission statement will close the workshop.

09.00 – 09.15	Welcome and Introduction
09.15 – 10.15	Impulse Presentations I
10.15 – 10.45	*Coffee Break*
10.45 – 11.45	Impulse Presentations II
11.45 – 13.00	*Lunch Break*
13.00 – 14.00	Impulse Presentations III
14.00 – 15.00	Small-group Discussion
15.00 – 16.00	Consolidation and Wrap-up

Table I. Preliminary workshop schedule.

After the workshop

All accepted workshop papers along with additional material such as presentation slides, results and meaningful photographs or video material will be published on the workshop website. Workshop participants will be asked if they are interested in jointly authoring a position paper or opinion piece about the essential workshop results and backgrounds.

Backgrounds of Organizers

Thomas Neumayr is an assistant professor and researcher at the University of Applied Sciences Upper Austria in Hagenberg and earned a PhD at the Johannes Kepler University Linz on description and analysis of hybrid collaboration. His research interests include collaborative practices of smaller-sized teams, interaction design, and personalization. He holds a BA and MA in Communication and Knowledge Media from the University of Applied Sciences Upper Austria. Thomas is the elected speaker of the special interest group ABIS of the German Gesellschaft für Informatik. He, along with Mirjam Augstein, received a best paper award for their Domino framework at ACM CSCW 2018 and was the main organizer together with Banu Saatçi of the first international workshop on hybrid collaboration in 2019.

Frederik Hirschmann is a researcher at the University of Applied Sciences Upper Austria in Hagenberg and a PhD student at the Johannes Kepler University Linz on the topic of hybrid collaboration spaces, i.e., the environments in which hybrid collaboration occurs. He holds an MSc in Human-Centered Computing from the University of Applied Sciences Upper Austria and one in Human-Computer Interaction from a joint degree program of the Paris Lodron Universität Salzburg and Salzburg University of Applied Sciences.

Mirjam Augstein is Professor for Personalized and Collaborative Systems at the University of Applied Sciences Upper Austria in Hagenberg. She received her PhD and recently her venia docendi at the Johannes Kepler University in Linz. Her main research interests include adaptive systems and computer-supported cooperative work. Further, Mirjam is interested in User Experience and Interaction Design as well as novel User Interfaces. She has recently coordinated several related research projects and organized several workshops in the area of HCI, e.g., the Workshop on Intelligent and Personalized Human-Computer Interaction series (associated with the HCI group of the German Computer Science Society) and has edited a book on Personalized HCI. Mirjam has been serving on the PC or as a reviewer at several top-tier ACM conferences in the area of HCI (e.g., CHI, CSCW, ISS, ICMI, TEI or DIS).

Johannes Schönböck is Professor for Social and Semantic Web at the University of Applied Sciences Upper Austria, Hagenberg campus. He earned a PhD from Johannes Kepler University Linz and Technical University Vienna with the greatest honors obtainable in the federal republic of Austria *sub auspiciis Praesidentis*.

Tuire Oittinen is a Senior Lecturer at the University of Jyväskylä. She received her PhD from the University of Jyväskylä, focusing on the coordination of actions in technology-mediated business meetings. In her current research, she uses multimodal conversation analysis (CA) to study human social conduct in authentic technology-mediated and hybrid collaborative settings. Tuire is the principal investigator of the project 'Participation and collaboration in hybrid interactional ecologies of learning', and she is the co-editor of a book 'Video-mediated L2 interaction: Conversation analytic research'.

Minttu Vänttinen is a postdoctoral researcher at the University of Jyväskylä. She received her PhD from the University of Jyväskylä, where her research focused on interaction around and via technology in face-to-face and hybrid educational settings. She uses multimodal conversation analysis (CA) and other qualitative methods to study technologized interaction, digital game-based learning, and collaboration in hybrid spaces.

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